

BACOU, Melanie

Applied Economist, Data Scientist, Geospatial Analyst

✉ [redacted] • 📞 [redacted] • 📠 [redacted]
📧 t.me/mbacou • 📞 wa.me/mbacou



Economist with 20+ years in public-sector consulting. Led economic development programs and research grants in the \$500K range. Advised government agencies and multilateral donors with large-scale data collection (field-based, mobile, and remote sensing methods), household survey design and analysis, market segmentation, agricultural value-chain analysis, rural finance, climate risk assessment and risk transfer.

I bring in:

- Strong quantitative, analytical, data modeling, and project management skills acquired in collaboration with top-rank economic research institutes and consulting firms
- A multidisciplinary background with ability to fundraise and liaise efficiently with investors, government officials, private-sector partners, technical experts, researchers, and beneficiaries
- Extensive work experience in fragile states (Mid-East, West/sub-Saharan Africa)

Consulting rate: **US \$650** / day

French (EU) / U.S. citizenships

🔗 github.com/mbacou
🌐 linkedin.com/in/mbacou
☎ 0000-0003-1810-6818

Services

Cost/benefit analysis
Economic impact evaluation
Spatial econometrics
Climate risk assessment
Business analytics
Data / map visualization

Languages

French (native)
English (bilingual)
German (limited working)
Russian (beginner)

Certifications

AWS Certified Data Engineer
SAS Programming III

TECH SKILLS

Statistics R, STATA, SAS, GAMS / LaTeX, Quarto, Typst

Data viz Shiny, Tableau®, Highcharts.js, D3.js, Superset, Grafana

GIS QGIS, GRASS, GDAL, GeoServer, Carto, Mapnik, Leaflet, Kepler.gl, Google Earth Engine

Databases MS SQL Server, MySQL, PostGIS, MongoDB, MariaDB, Elasticsearch, DuckDB

DevOps Linux (Ubuntu, Arch), Git, CI/CD, Docker, AWS cloud deployment, containerization, orchestration (Airflow)

Survey ODK, CSPro, SurveyCTO, CommCare, Fulcrum, GISCloud

Agile development Jira, Trello, Asana, Taiga.io, ClubHouse

EXPERIENCE



2021 Aug - 2022 Jan
6 mos

Porto, Portugal
(remote)

International Water Management Institute

hydrology

Consultant - Data Modeling and Visualization

Built advanced data pipelines (ETL) and interactive tools to track changes in water budgets and water productivity across multiple river basins. Part of FAO Water Secure Africa initiative (WASA) to inform equitable and sustainable water use policies.

R Shiny AWS Docker Highcharts.js D3.js



WorldCover PBC

insurtech, agrometeorology

Head of Research

2017 - 2021
4 yrs

Washington, DC
Ghana
Uganda
Togo
Cameroon

WorldCover engineers solutions to better design, target, and deliver affordable (parametric) weather insurance to small-scale farmers (Ghana, Togo, Kenya, Uganda), and to large-scale fruit and nut tree growers, and to protect entire agricultural value chains against climate risks. Built insurance analytics infra and team. Led agrometeorological research across 20 value-chains and agroclimatic zones. Rolled out a real-time pricing engine and multiple APIs on AWS cloud. Technical platform acquired in April 2021.

R Python Shiny AWS PostgreSQL quarto airflow fulcrum ClubHouse



Gates Foundation

agricultural research

Consultant - Agricultural Economist

2017 Apr - Dec
9 mos

Washington, DC
Seattle, WA

Business-oriented typologies and characterization of small-holder farming systems in Ghana, Ethiopia, Tanzania, and Nigeria. Under a work assignment with Spatial Development International (now TRC Companies Inc).

R STATA QGIS



Abt Associates

agricultural research

Consultant - GIS Analyst

2017 Jun - Jul
2 mos

Washington, DC

Geospatial analysis to inform USAID AgResults program evaluation in Nigeria, Kenya, and Uganda.

R STATA QGIS



International Food Policy Research Institute

economic research

Snr. Program Manager

2011 - 2016
6 yrs 8 mos

Washington, DC
Nairobi, Kenya
Addis, Ethiopia

Generated subnational socio-economic and biophysical datasets to inform and upscale agricultural research investments in sub-Saharan Africa, track adoption of new agricultural technologies, track CGIAR research investments around the globe, and monitor the impact of policies and investments on small farm-holder livelihoods.

Geospatial Analyst, HarvestChoice Program

2010 - 2014
4 yrs 4 mos

Research and analytics for the Global Development Program of the Gates Foundation, Feed the Future Africa RISING, the G-8 New Alliance for Food Security and Nutrition, the Alliance for a Green Revolution in Africa (AGRA), and several CGIAR programs focused on agricultural productivity improvement, food security, and poverty reduction in sub-Saharan Africa. In charge of HarvestChoice model, platform, and data releases.

Instructor / CMS Web Design, AGRODEP Program

2011 May - Oct
6 mos

Dakar, Senegal

Project management duties for the African Growth and Development Policy Modeling Consortium (AGRODEP). Website migration to Drupal CMS, incl. preparation and review of design mockups, solution architecture, migration of existing content, training of AGRODEP staff members. Conducted a 3-day training to AGRODEP awardees on household survey data sampling and processing in Dakar, Senegal. AGRODEP is a program sponsored in part by USAID and UKAid.

R STATA Tableau® ArcGIS QGIS CKAN Drupal Wordpress



Gates Foundation

agricultural research

Consultant - Agricultural Economist

2012 - 2014
2 yrs

Washington, DC
Seattle, WA

Country diagnostics, farmer segmentation, and data visualization for the Foundation's Agricultural Development Program. Under a work assignment with Spatial Development International (now **TRC Companies Inc.**).

R STATA QGIS



African Development Bank

international development

Team Lead - Procurement ERP

2008 Oct - 2009 Jan
4 mos

Tunis, Tunisia

Led a 4-month ICT project for the Bank's procurement and fiduciary services division aimed at a complete upgrade of its internal SAP® modules (procurement workflow and reporting tools) and external Web CMS (procurement tools and information for borrowing governments and business partners). Work assignment with **Crown Agents UK**.

IIS MS FrontPage SAP®



Development Gateway Foundation

aid procurement

Consultant - Monitoring & Evaluation of ICT Programs

2008 May - Sep
5 mos

Washington, DC

Helped conduct a 5-month evaluation of the relevance, effectiveness and impact, sustainability, governance and management structure of DGF (5 international programs and staff of 84). The evaluation was forward-looking and provided substantial strategic recommendations to DGF in the areas of ICT4D and aid effectiveness. Data collection included desk review, on-line surveys, usage analysis of DGF servers, phone and in-person stakeholder interviews with over 45 individuals. Work assignment with **AMEX International**.



Abt Associates

public policy, health, agriculture

Senior Analyst

2005 - 2007
2 yrs

Washington, DC
Sanaa, Yemen

Managed environmental regulatory cost-benefit analyses and program evaluations for the **U.S. EPA** Office of Air Quality and Office of Pesticide Programs (multi-year \$200K - \$500K task orders).

Advised and led ICT solutions for **USAID**, **MCC**, and **WTO**-funded projects. Supervised analysts, programmers, graphic designers, and content experts:

- USAID Global Business, Trade & Investment Services IQC. <https://gbti2.info/>.
- USAID Health Systems 20/20, Yemen. Advised the Yemen Ministry of Health on data policy and strategy. Built first agency website. <https://moh.gov.ye/>.
- Health Systems Action Network (HSAN) <https://hsanet.org/>.
- U.S. EPA Office of Emergency Management (OEM). Provided 8-month support to the Office Evaluation and Communication Division for its website development and information design. <https://epa.gov/emergencies/>

SQL STATA ArcGIS



2001 - 2005
4 yrs

West Lafayette, IN

Center for Global Trade Analysis

economic research

ICT and Data Specialist

- Built and administered web (IIS 5) and database servers (SQL Server 2000) in support of the GTAP Network (over 8,000 researchers in 110 countries).
- Ensured efficient communication and on-time data delivery to researchers in the Network
- Carried out vendor reviews and selections and supervised all aspects of web application rollouts (Visual Basic .Net, C#, JScript)
- Built web applications to streamline workshop and conference planning. Web app still fully operational since 2005.
- Coordinated all distance learning activities
- Screened funding opportunities, prepared project proposals and progress reports

VB.Net IIS MS SQL Server MS FrontPage Dreamweaver

Fireworks Audio/Video editing



2000 Jan - Aug
8 mos

Den Haag
The Netherlands

Landbouw Economisch Instituut - Wageningen UR

agricultural research

Information Management Specialist - e-learning

- Planned and coordinated a first on-line training program on horticultural systems between WUR and agricultural academies in China
- Delivered initial training of trainers

SQL BlackBoard LMS ArcGIS



1998 - 1999
1 yr 4 mos

West Lafayette, IN
Harare, Zimbabwe

Purdue University

trade economics

Research Assistant - Instructional design, web application development

- Technical expert for a 6-month distance learning project for the Zimbabwean Tariff Commission sponsored by the **World Bank**
- Developed a distance learning course on general equilibrium theories and applications, sponsored by the United Nations Economic Commission for Africa
- Fully planned, programmed and designed all educational websites, multimedia content, and CD-ROMs.

Linux VCS Fortran GAMS GEMPack C# Moodle LMS MS SQL Server IIS

MS FrontPage Dreamweaver Fireworks Audio/Video editing



1996 Jan - May
5 mos

London, UK

SGEAM (SG Emerging Europe Asset Management)

private equity

Marketing and Technical Assistant

- Assisted the launch of high-risk investment funds on Eastern European markets
- Performed ratio analysis and financial reporting (translation of Polish and Czech accounting standards into IAS)
- Built financial databases and automated reports to assist market analysts with quarterly stock performance monitoring

MSAccess MSEXcel DB2 Bloomberg



1995 Jul - Oct
4 mos

Paris, France

FIMATEX (Boursorama Bank)

derivatives trading

Marketing and Sales Assistant

- Took part in a strategic diversification from commodity to index derivatives markets
- Negotiated new marketing partnerships with the financial press and software vendors
- Collaborated with traders to produce daily market reports on French MATIF and MONEP bond and index futures and options
- Designed first corporate website [fimatex.fr](https://fr.wikipedia.org/wiki/Fimatex), see <https://fr.wikipedia.org/wiki/Fimatex> (now Boursorama)

Visual Basic Lotus Notes DB2 MSExcel



1995 Jan - Apr
4 mos

Cergy Pontoise,
France

SEAT France

automotive

Marketing and Sales Intern

Contributed to quarterly promotional campaigns with advertising agencies and tracked all merchandising programs to 100+ SEAT car dealers across France.

DB2 MSExcel

EDUCATION



West Lafayette, IN
United States

Purdue University | M.Sc. Agricultural Economics

1997 - 2001

🏆 Award of Merit, Outstanding M.Sc. Thesis

Thesis: *Economy-wide Effects of Climate Variability and Climate Prediction in Mozambique*, under the supervision of Dr. Channing Arndt.

Activities: started and taught Purdue Argentine Tango student organization.



Paris
Oxford
Berlin

ESCP Europe (EAP) | M.A. International Management

1994 - 1997

🏆 Diplôme de Grande École de Commerce, Diplom-Kaufmann

Previously EAP, 3-year / 3-country program, among France's top 3 business degrees.

Thesis: *Agricultural Derivatives Markets – A Substitute to the Common Agricultural Policy Support Price Mechanisms*.



Paris, France

Lycée Louis-le-Grand | Classe Prépa HEC

1992 - 1994

2-year intensive B.Sc. degree equivalent.

Business mathematics, statistics, economics theory, contemporary history, philosophy.



Saint Petersburg

Herzen State University | Russian as foreign language

2023 Jun - Sep

🏆 Grant recipient, Ministry of Education

Intensive summer course in Russian language, culture, and literature.

PUBLICATIONS

<i>Poverty, Land, and Climate in Africa South of the Sahara: an empirical analysis</i> . International Food Policy Research Institute. Presented at CSAE 2016	Dec 2016	↓
<i>CELL5M: A Geospatial Database of Agricultural Indicators for Africa South of the Sahara</i> , F1000Research 2016, 5:2490.	Oct 2016	↓
<i>Subnational Socio-Economic Dataset Availability</i> , Nature Climate Change 6:115-116. DOI: 10.1038/nclimate2842	Jan 2016	↓
<i>Characterization of Smallholder Households in Ethiopia - A household level summary of the context of smallholder agricultural production</i> , HarvestChoice, International Food Policy Research Institute.	Aug 2010	↓
<i>Evaluation of the Development Gateway Foundation, FY 2005-2007</i> , with Marc Shapiro, Christine Davachi, David Witzel. Washington, DC, October 2008.	Oct 2008	↓
<i>Technical Report on Ozone Exposure, Risk, and Impact Assessments for Vegetation</i> , Prepared for the Office of Air Quality Planning and Standards, U.S. Environmental Protection Agency.	Jan 2007	↓
<i>Climate Forecasts in Mozambique: an Economic Perspective</i> , in Karen L. O'Brien and Vogel, Coleen H. (eds.) Climate Forecasts in Southern Africa. Aldershot, U.K. Ashgate.	Apr 2005	↓
<i>Economy-Wide Effects of Climate Variability and Climate Prediction in Mozambique</i> , American Journal of Agricultural Economics 82 (3) 750-754.	Aug 2000	↓